# Scalability & Future Plan

Phase 8 | Roadmap and growth considerations

## Current foundation

The current architecture separates the frontend, API, AI generation, and persistence services and includes Docker and cloud-hosting configuration. This provides a starting point for further work; it does not establish measured scalability or production capacity.

## Proposed roadmap

| Horizon | Candidate work | Reason |
|---|---|---|
| Near term | Date validation, frontend/end-to-end tests, structured error handling, usage and latency logging. | Improve correctness and release confidence. |
| Next | Explicit public-sharing permissions; account recovery and email verification. | Clarify access and complete account lifecycle. |
| Later | Maps, transit, weather, opening-hour or verified price integrations. | Add context with provenance and freshness. |
| Later | Collaborative editing, preference profiles, recommendation feedback. | Support repeat use and group planning after validating demand. |

## Scale risks to address

- Set provider timeouts, rate limits, and cost controls before significant traffic.
- Measure generation latency and fallback rate by provider.
- Review database indexes, connection behavior, and ownership/RLS under expected load.
- Avoid relying on process memory for multi-instance deployment.
- Define observability and incident handling for API, provider, and database failures.

## Roadmap rule

*Prioritize future features using traveler feedback and measured use. The items above are proposals, not committed delivery dates or existing capabilities.*

Prepared from the current project source and configuration. Product assumptions and proposed work are labeled; no user research, test result, or deployment result is implied unless explicitly stated.